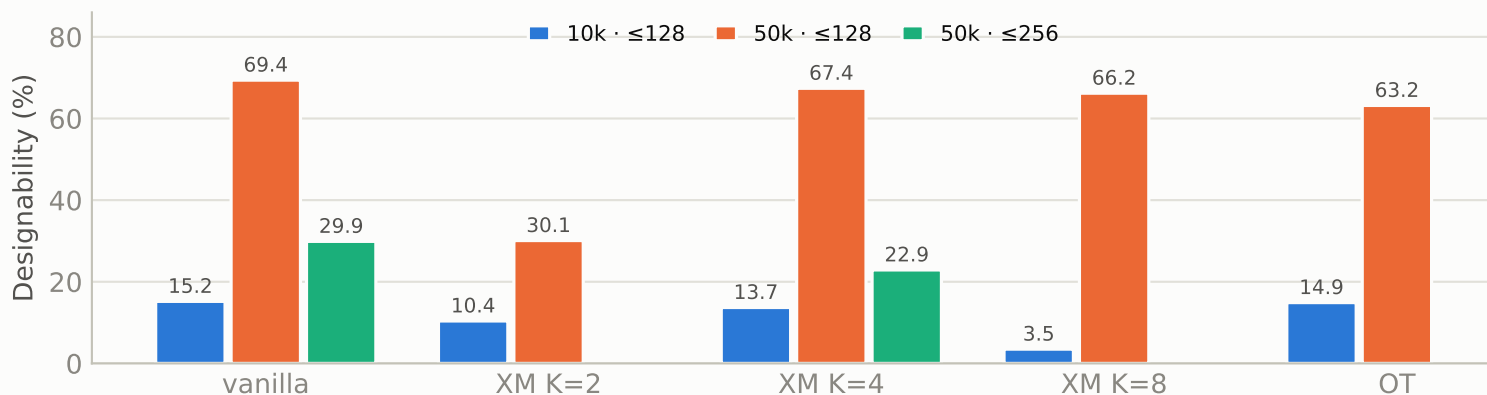


Explorative Modeling (XM) for FrameFlow

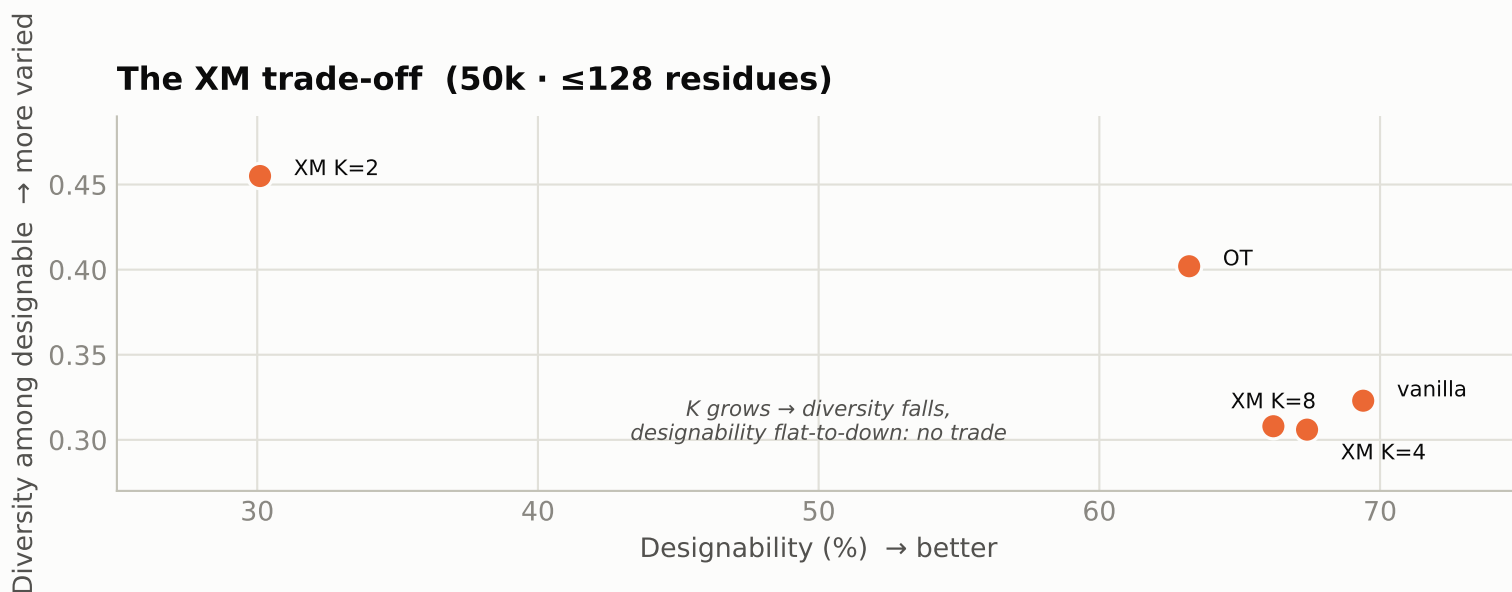
Best-of-K coupling selection vs. plain independent coupling — designability across four training regimes.

Verdict: across every regime tested — 10k & 50k steps, ≤ 128 & ≤ 256 residues — plain vanilla independent coupling wins. XM never beats it; larger K trades structural diversity for no designability gain, and at ≤ 256 XM is clearly worse. The null result is not a small-protein artifact. Experiment complete.

Designability by arm and regime



The XM trade-off (50k · ≤ 128 residues)



All numbers

Arm	10k ≤ 128	50k ≤ 128	50k ≤ 256	Div(d) ≤ 128	Div(d) ≤ 256	Nov ≤ 128
vanilla	15.2%	69.4%	29.9%	0.323	0.733	0.197
XM K=2	10.4%	30.1%	—	0.455	—	0.215
XM K=4	13.7%	67.4%	22.9%	0.306	0.674	0.203
XM K=8	3.5%	66.2%	—	0.308	—	0.205
OT	14.9%	63.2%	—	0.402	—	0.227